

Yijing Wu

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SKILLS

Programming Languages: Python, JavaScript, TypeScript, Bash, SQL, Java, HTML, CSS, Swift, C/C++
Frameworks: React, React Native, Node.js, Django, Express, Next.js, Vue.js, Redux, GraphQL
Cloud and DevOps: AWS (Lambda, ECR, DynamoDB, EC2, VPC, IAM), Azure, Terraform, Jenkins, Docker, Linux
Others: MongoDB, Firebase, MySQL, Git, Open AI, CocoaPods, Restful API, PyTest, Unittest, ts-Jest, Jupyter Notebook

EXPERIENCE

Autodesk San Francisco, CA (Remote)
Cloud DevOps Engineer Intern May 2023 – Present

- Developed a Serverless Slack bot in **Python**, integrating ServiceNow and JIRA to streamline ticket management via Slack APIs and automate issue resolution by Generative AI techniques, resulting in a significant reduction in support response times
- Utilized **Azure OpenAI LLM** and **Gen AI** techniques, applying **prompt engineering**, **model inference**, **embeddings** and **NLP** for interactive conversations on custom data, including document search, API calling, and data plotting functionalities
- Established a robust **CI/CD** pipeline using **Terraform** and **Jenkins**, following Infrastructure as Code (**IaC**) principles to enable end-to-end cloud deployment automation, thereby enhancing efficiency and maintaining consistency
- Utilized **AWS Serverless** architecture, employing services such as **Lambda**, **ECR**, **S3** Bucket, **DynamoDB**, **SageMaker**, **VPC**, **API Gateway**, **IAM**, **CloudWatch**, and **KMS** to optimize the backend infrastructure for seamless operations
- Trained a multi-class classification model with **sklearn** and **Logistic Regression**, achieved an 81% accuracy in issue diagnosis

Upduo San Francisco, CA (Remote)
Full-Stack Software Engineer Intern June 2022 – December 2022

- Contributed as a full-stack engineer in an agile team to build, enhance and maintain cross-platform web & mobile application
- Collaborated with product and design team, conducted front-end development in **React** and **React Native** along with **Redux** through **TypeScript** to enhance UI/UX and build graph data visualization, animations, and kudos interface
- Worked on **Django** backend (Python), applied **Celery** to manage asynchronous tasks effectively, optimized **GraphQL** to minimize queries to **PostgreSQL** database, created **Unit Testing** and **Integration Testing** to ensure application's robustness
- Integrated the **iOS** framework **CallKit** and **Android** **ConnectionService**, utilized **Objective-C** in **Xcode** and **Java** in **Android Studio** to implement a video call notification feature, which led to a significant enhancement in match pick-up rate
- Performed camera and microphone selection and checking features through **Agora** to ensure **WebRTC** video calling quality

Huawei Technologies Shanghai, China
Algorithm Engineer Intern July 2020 – September 2020

- Applied **computer vision** techniques to calibrate and correct camera sensor distortion in self-driving vehicles' video streams
- Implemented cylindrical projection and single-point mapping in **Python**, resulting in reduction of distortion correction errors from 1.395 ± 0.136 pixels to 0.055 ± 0.192 pixels
- Developed mapping tables and leveraging **OpenCV** video processing algorithms in **C** to enhance DSP processing efficiency

PROJECTS

Rubik's Cube Helper November 2021 - November 2022

- Created a dynamic app with **React (JavaScript)** for web, **Swift** and **CocoaPods** for **iOS** to customize Cube experience
- Implemented responsive design, accessibility development, and multiple interactions to meet WCAG 2.1 standard on web
- Delivered **Node.js** backend with **Express**, hosted on **AWS EC2**, and integrated **MongoDB** to facilitate data storage, enabling real-time tracking of world records and enhanced user engagement

EEG-based Brain-Computer Interface Research September 2019 – June 2021

- Constructed classification models of brain for material-specific Sternberg task based on 128-channel EEG dataset
- Extracted EEG features using empirical mode decomposition (EMD) and phase space reconstruction (PSR) in Matlab
- Employed machine learning such as SVM, KNN, Random Forest (RF) with nested cross-validation in Python
- Improved classification performance from 71.45% to 91.82% and published results as first author in an IEEE paper

EDUCATION

Northeastern University Seattle, WA
Master of Science in Information Systems; GPA: 3.95/4.0 September 2021 – December 2023

Tongji University Shanghai, China
Bachelor of Engineering in Communications Engineering September 2017 – June 2021

ACADEMIC PUBLICATION

Classification of EEG signals during Working-Memory Maintenance based on Phase Space Reconstruction of Empirical Mode Decomposition, Oct 2020, ISBN:978-1-6654-2299-4